

Loan Portfolio & Risk Analysis System

End-to-End Database Analytics, Credit Risk Categorization & Management Querying

Domain: Banking & Financial Analytics

Tooling: SQL (MySQL / PostgreSQL / ANSI SQL)

Project Type: GitHub Professional Repository

Queries: 80 Production-Ready Solutions

Executive Overview & Business Objective

This repository contains a comprehensive SQL analysis framework designed for financial institutions to evaluate loan portfolio performance, quantify credit risk exposure, monitor repayment status, and calculate key strategic KPIs. The dataset analyzes customer demographics, loan distributions, default frequencies, and recovery rates to assist leadership in data-driven lending decision-making.

DATABASE SCHEMA: LOAN TABLE ATTRIBUTES

Loan_ID	Customer_ID	Customer_Name	Gender	Age	City	State	Employment_Type
Annual_Income	Loan_Type	Loan_Amount	Interest_Rate	Loan_Term	Application_Date		
Approval_Date	Disbursement_Date	Loan_Status	Credit_Score	Monthly_EMI	Amount_Paid		
Outstanding_Amount	Overdue_Amount	Payment_Status					

1. Basic SQL Analysis

QUESTION 01

Display all loan records from the database.

```
SELECT *  
FROM loan;
```

QUESTION 02

Display only customer name, city, loan type, and loan amount.

```
SELECT Customer_Name, City, Loan_Type, Loan_Amount  
FROM loan;
```

QUESTION 03

Find all customers who have applied for a Personal Loan.

```
SELECT Customer_Name
FROM loan
WHERE Loan_Type = 'Personal';
```

QUESTION 04

Find customers whose loan amount is greater than ₹5,00,000.

```
SELECT Customer_Name, Loan_Amount
FROM loan
WHERE Loan_Amount > 500000;
```

QUESTION 05

Find customers from a particular city (e.g., Delhi).

```
SELECT Customer_Name, City
FROM loan
WHERE City = 'Delhi';
```

QUESTION 06

Display customers whose credit score is greater than 750.

```
SELECT Customer_Name, Credit_Score
FROM loan
WHERE Credit_Score > 750;
```

QUESTION 07

Find customers whose annual income is between ₹5 lakh and ₹15 lakh.

```
SELECT Customer_Name, Annual_Income
FROM loan
WHERE Annual_Income BETWEEN 500000 AND 1500000;
```

QUESTION 08

Display all customers whose loan status is Approved.

```
SELECT Customer_Name, Loan_Status
FROM loan
WHERE Loan_Status = 'Approved';
```

QUESTION 09

Find all customers whose payment status is Overdue.

```
SELECT Customer_Name, Payment_Status
FROM loan
WHERE Payment_Status = 'Overdue';
```

QUESTION 10

Display loans where outstanding amount is greater than ₹2,00,000.

```
SELECT Customer_Name, Outstanding_Amount
FROM loan
WHERE Outstanding_Amount > 200000;
```

2. Filtering & Sorting

QUESTION 11

Display the 10 largest loans.

```
SELECT Customer_Name, Loan_Amount
FROM loan
ORDER BY Loan_Amount DESC
LIMIT 10;
```

QUESTION 12

Display customers in descending order of credit score.

```
SELECT Customer_Name, Credit_Score
FROM loan
ORDER BY Credit_Score DESC;
```

QUESTION 13

Display customers in ascending order of loan amount.

```
SELECT Customer_Name, Loan_Amount
FROM loan
ORDER BY Loan_Amount ASC;
```

QUESTION 14

Find loans with interest rates greater than 10%.

```
SELECT Customer_Name, Interest_Rate, Loan_Type
FROM loan
WHERE Interest_Rate > 10;
```

QUESTION 15

Find customers between the ages of 25 and 40.

```
SELECT Customer_Name, Age
FROM loan
WHERE Age BETWEEN 25 AND 40
ORDER BY Age ASC;
```

QUESTION 16

Find customers whose names start with 'A'.

```
SELECT Customer_Name
FROM loan
WHERE Customer_Name LIKE 'A%';
```

QUESTION 17

Find customers whose city contains the letter 'a'.

```
SELECT Customer_Name, City
FROM loan
WHERE City LIKE '%a%';
```

QUESTION 18

Find customers who are not from a particular state (e.g., 'UP').

```
SELECT Customer_Name, State
FROM loan
WHERE State <> 'UP';
```

QUESTION 19

Find loans that are either Approved or Disbursed.

```
SELECT Customer_Name, Loan_Amount, Loan_Status
FROM loan
WHERE Loan_Status IN ('Approved', 'Disbursed');
```

QUESTION 20

Find customers whose payment status is not Paid (e.g., Pending or Overdue).

```
SELECT Customer_Name, Payment_Status
FROM loan
WHERE Payment_Status <> 'Paid';
```

3. Aggregate Analysis

QUESTION 21

Find total number of loan applications.

```
SELECT COUNT(Loan_ID) AS total_loan_applications
FROM loan;
```

QUESTION 22

Find total number of approved loans.

```
SELECT COUNT(Loan_ID) AS approved_loans
FROM loan
WHERE Loan_Status = 'Approved';
```

QUESTION 23

Find total loan amount requested across all applications.

```
SELECT SUM(Loan_Amount) AS total_loan_requested
FROM loan;
```

QUESTION 24

Find total loan amount disbursed.

```
SELECT SUM(Loan_Amount) AS total_loan_disbursed
FROM loan
WHERE Loan_Status = 'Disbursed';
```

QUESTION 25

Find total outstanding loan amount.

```
SELECT SUM(Outstanding_Amount) AS total_outstanding_amount
FROM loan;
```

QUESTION 26

Find total overdue amount.

```
SELECT SUM(Overdue_Amount) AS total_overdue_amount
FROM loan;
```

QUESTION 27

Find average loan amount.

```
SELECT AVG(Loan_Amount) AS avg_loan_amount
FROM loan;
```

QUESTION 28

Find average interest rate.

```
SELECT AVG(Interest_Rate) AS avg_interest_rate
FROM loan;
```

QUESTION 29

Find maximum loan amount.

```
SELECT MAX(Loan_Amount) AS max_loan_amount
FROM loan;
```

QUESTION 30

Find minimum loan amount.

```
SELECT MIN(Loan_Amount) AS min_loan_amount
FROM loan;
```

QUESTION 31

Find average credit score of borrowers.

```
SELECT AVG(Credit_Score) AS avg_credit_score
FROM loan;
```

QUESTION 32

Find average annual income of borrowers.

```
SELECT AVG(Annual_Income) AS avg_annual_income
FROM loan;
```

4. GROUP BY Analysis

QUESTION 33

Find the number of loans for each loan type.

```
SELECT Loan_Type, COUNT(*) AS loan_count
FROM loan
GROUP BY Loan_Type;
```

QUESTION 34

Find total loan amount for each loan type.

```
SELECT Loan_Type, SUM(Loan_Amount) AS total_loan_amount
FROM loan
GROUP BY Loan_Type;
```

QUESTION 35

Find average loan amount for each loan type.

```
SELECT Loan_Type, AVG(Loan_Amount) AS avg_loan_amount
FROM loan
GROUP BY Loan_Type;
```

QUESTION 36

Find total loan amount by city.

```
SELECT City, SUM(Loan_Amount) AS total_loan_amount
FROM loan
GROUP BY City;
```

QUESTION 37

Find number of customers by state.

```
SELECT State, COUNT(Customer_Name) AS customer_count
FROM loan
GROUP BY State;
```

QUESTION 38

Find average credit score by loan type.

```
SELECT Loan_Type, AVG(Credit_Score) AS avg_credit_score
FROM loan
GROUP BY Loan_Type;
```

QUESTION 39

Find total outstanding amount by loan type.

```
SELECT Loan_Type, SUM(Outstanding_Amount) AS total_outstanding
FROM loan
GROUP BY Loan_Type;
```

QUESTION 40

Find total overdue amount by city.

```
SELECT City, SUM(Overdue_Amount) AS total_overdue
FROM loan
GROUP BY City;
```

QUESTION 41

Find the number of loans by payment status.

```
SELECT Payment_Status, COUNT(*) AS loan_count
FROM loan
GROUP BY Payment_Status;
```

QUESTION 42

Find the number of loans by employment type.

```
SELECT Employment_Type, COUNT(*) AS borrower_count
FROM loan
GROUP BY Employment_Type;
```

5. HAVING Clause

QUESTION 43

Find loan types having more than 5 loans.

```
SELECT Loan_Type, COUNT(*) AS loan_count
FROM loan
GROUP BY Loan_Type
HAVING COUNT(*) > 5;
```

QUESTION 44

Find cities where total loan disbursement is greater than ₹50 lakh.

```
SELECT City, SUM(Loan_Amount) AS total_disbursement
FROM loan
GROUP BY City
HAVING SUM(Loan_Amount) > 5000000;
```

QUESTION 45

Find loan types whose average loan amount is greater than ₹3 lakh.

```
SELECT Loan_Type, AVG(Loan_Amount) AS avg_loan_amount
FROM loan
GROUP BY Loan_Type
HAVING AVG(Loan_Amount) > 300000;
```

QUESTION 46

Find cities having total overdue amount greater than ₹1 lakh.

```
SELECT City, SUM(Overdue_Amount) AS total_overdue
FROM loan
GROUP BY City
HAVING SUM(Overdue_Amount) > 100000;
```

QUESTION 47

Find employment types having more than 10 borrowers.

```
SELECT Employment_Type, COUNT(Customer_Name) AS borrower_count
FROM loan
GROUP BY Employment_Type
HAVING COUNT(Customer_Name) > 10;
```

6. CASE Statement — Risk Classification

QUESTION 48

Categorize customers into Risk Categories based on Credit Score.

```
SELECT Customer_Name, Credit_Score,
       CASE
         WHEN Credit_Score >= 750 THEN 'Low Risk'
         WHEN Credit_Score BETWEEN 650 AND 749 THEN 'Medium Risk'
         WHEN Credit_Score BETWEEN 550 AND 649 THEN 'High Risk'
         ELSE 'Very High Risk'
       END AS Risk_Category
FROM loan;
```

QUESTION 49

Count the number of customers in each risk category.

```
SELECT
  CASE
    WHEN Credit_Score >= 750 THEN 'Low Risk'
    WHEN Credit_Score BETWEEN 650 AND 749 THEN 'Medium Risk'
    WHEN Credit_Score BETWEEN 550 AND 649 THEN 'High Risk'
    ELSE 'Very High Risk'
  END AS Risk_Category,
  COUNT(*) AS customer_count
FROM loan
GROUP BY Risk_Category;
```

QUESTION 50

Find total outstanding amount for each risk category.

```
SELECT
  CASE
    WHEN Credit_Score >= 750 THEN 'Low Risk'
    WHEN Credit_Score BETWEEN 650 AND 749 THEN 'Medium Risk'
    WHEN Credit_Score BETWEEN 550 AND 649 THEN 'High Risk'
    ELSE 'Very High Risk'
  END AS Risk_Category,
  SUM(Outstanding_Amount) AS total_outstanding
FROM loan
GROUP BY Risk_Category;
```

QUESTION 51

Find total overdue amount for each risk category.

```
SELECT
    CASE
        WHEN Credit_Score >= 750 THEN 'Low Risk'
        WHEN Credit_Score BETWEEN 650 AND 749 THEN 'Medium Risk'
        WHEN Credit_Score BETWEEN 550 AND 649 THEN 'High Risk'
        ELSE 'Very High Risk'
    END AS Risk_Category,
    SUM(Overdue_Amount) AS total_overdue
FROM loan
GROUP BY Risk_Category;
```

7. Business KPI Analysis

QUESTION 52

Calculate Loan Approval Rate (%).

```
SELECT
    SUM(CASE WHEN Loan_Status = 'Approved' THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS
    approval_rate_pct
FROM loan;
```

QUESTION 53

Calculate Loan Rejection Rate (%).

```
SELECT
    SUM(CASE WHEN Loan_Status = 'Rejected' THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS
    rejection_rate_pct
FROM loan;
```

QUESTION 54

Calculate Average Loan Amount KPI.

```
SELECT AVG(Loan_Amount) AS avg_loan_amount
FROM loan;
```

QUESTION 55

Calculate Total Disbursed Amount KPI.

```
SELECT SUM(Loan_Amount) AS total_disbursed_amount
FROM loan
WHERE Loan_Status = 'Disbursed';
```

QUESTION 56

Calculate Total Outstanding Amount KPI.

```
SELECT SUM(Outstanding_Amount) AS total_outstanding_amount
FROM loan;
```

QUESTION 57

Calculate Total Overdue Amount KPI.

```
SELECT SUM(Overdue_Amount) AS total_overdue_amount
FROM loan;
```

QUESTION 58

Calculate Average Credit Score KPI.

```
SELECT AVG(Credit_Score) AS avg_credit_score
FROM loan;
```

QUESTION 59

Find Default Customer Count.

```
SELECT COUNT(*) AS default_customer_count
FROM loan
WHERE Payment_Status = 'Default';
```

QUESTION 60

Calculate Percentage of Overdue Loans.

```
SELECT
    SUM(CASE WHEN Payment_Status = 'Overdue' THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS
    overdue_loan_pct
FROM loan;
```

8. Date Analysis

QUESTION 61

Find loans applied for in a particular year (e.g., 2026).

```
SELECT Customer_Name, Application_Date
FROM loan
WHERE YEAR(Application_Date) = 2026;
```

QUESTION 62

Find loans approved during a particular month (e.g., January).

```
SELECT Customer_Name, Application_Date
FROM loan
WHERE MONTH(Application_Date) = 1;
```

QUESTION 63

Find the number of loans applied for each year.

```
SELECT YEAR(Application_Date) AS app_year, COUNT(*) AS application_count
FROM loan
GROUP BY YEAR(Application_Date);
```

QUESTION 64

Find the total loan amount disbursed each year.

```
SELECT YEAR(Disbursement_Date) AS disb_year, SUM(Loan_Amount) AS total_disbursed
FROM loan
GROUP BY YEAR(Disbursement_Date);
```

QUESTION 65

Find customers whose loan was approved but not yet disbursed.

```
SELECT Customer_Name, Loan_Status
FROM loan
WHERE Loan_Status = 'Approved' AND Disbursement_Date IS NULL;
```

QUESTION 66

Find the number of loans approved in each month.

```
SELECT MONTH(Application_Date) AS app_month, COUNT(*) AS loan_count
FROM loan
GROUP BY MONTH(Application_Date);
```

QUESTION 67

Find the average number of days between application and approval.

```
SELECT AVG(DATEDIFF(Approval_Date, Application_Date)) AS avg_approval_days
FROM loan;
```

9. Subquery Challenges

QUESTION 68

Find customers whose loan amount is greater than the average loan amount.

```
SELECT Customer_Name, Loan_Amount
FROM loan
WHERE Loan_Amount > (SELECT AVG(Loan_Amount) FROM loan);
```

QUESTION 69

Find customers whose credit score is greater than the average credit score.

```
SELECT Customer_Name, Credit_Score
FROM loan
WHERE Credit_Score > (SELECT AVG(Credit_Score) FROM loan);
```

QUESTION 70

Find the customer who has taken the highest loan amount.

```
SELECT Customer_Name, Loan_Amount
FROM loan
WHERE Loan_Amount = (SELECT MAX(Loan_Amount) FROM loan);
```

QUESTION 71

Find customers who have outstanding amounts greater than the average outstanding amount.

```
SELECT *
FROM loan
WHERE Outstanding_Amount > (SELECT AVG(Outstanding_Amount) FROM loan);
```

QUESTION 72

Find the loan type having the highest total loan amount.

```
SELECT Loan_Type
FROM loan
GROUP BY Loan_Type
HAVING SUM(Loan_Amount) = (
    SELECT MAX(total_sum)
    FROM (
        SELECT SUM(Loan_Amount) AS total_sum
        FROM loan
        GROUP BY Loan_Type
    ) AS sub
);
```

QUESTION 73

Find customers whose annual income is greater than the average annual income.

```
SELECT Customer_Name, Annual_Income
FROM loan
WHERE Annual_Income > (SELECT AVG(Annual_Income) FROM loan);
```

10. Advanced SQL — Top Management Questions

QUESTION 74

Find the top 5 customers by total loan amount.

```
SELECT Customer_Name, SUM(Loan_Amount) AS total_loan
FROM loan
GROUP BY Customer_Name
ORDER BY total_loan DESC
LIMIT 5;
```

QUESTION 75

Find the top 3 cities by total loan disbursement.

```
SELECT City, SUM(Loan_Amount) AS total_disbursement
FROM loan
GROUP BY City
ORDER BY total_disbursement DESC
LIMIT 3;
```

QUESTION 76

Find the top 3 loan types by outstanding amount.

```
SELECT Loan_Type, SUM(Outstanding_Amount) AS total_outstanding
FROM loan
GROUP BY Loan_Type
ORDER BY total_outstanding DESC
LIMIT 3;
```

QUESTION 77

Find customers who have Credit Score < 650 and Overdue Amount < ₹5,00,000.

```
SELECT Customer_Name, Credit_Score, Overdue_Amount
FROM loan
WHERE Credit_Score < 650 AND Overdue_Amount < 500000;
```

QUESTION 78

Identify customers whose loan amount is higher than their annual income.

```
SELECT Customer_Name, Loan_Amount, Annual_Income
FROM loan
WHERE Loan_Amount > Annual_Income;
```

QUESTION 79

Identify customers who have multiple loans.

```
SELECT Customer_ID, COUNT(*) AS total_loans
FROM loan
GROUP BY Customer_ID
HAVING COUNT(*) > 1;
```

QUESTION 80

Find customers whose previous loan is closed but who have another active loan.

```
SELECT DISTINCT a.Customer_ID
FROM loan a
JOIN loan b ON a.Customer_ID = b.Customer_ID
WHERE a.Loan_Status = 'Closed' AND b.Loan_Status = 'Active';
```